# Lesson 7: On the Move

Unreal Engine (optional advanced track)

VIDEO GAME CREATION

**This week: enemies stop standing still. Your job is to make every enemy advance slowly toward the player, and clean up any that reach the bottom of the screen without being shot.**

**Open your own [YourName]Game project from Lesson 6 - don't start a new one.**

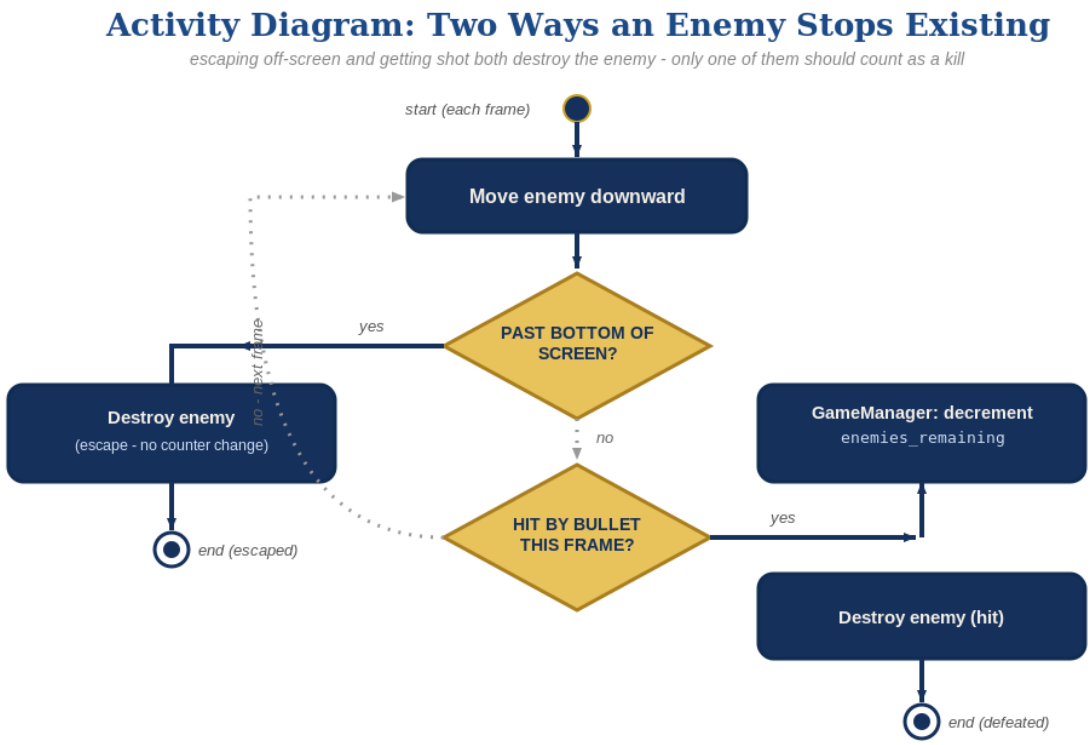
*Tip: no new assets this week - this is a logic-only lesson.*

## Setup

1. Open your [YourName]Game project from Lesson 6.

## Level 1 - Minimal (MVP)

*Every enemy moves slowly downward, toward the player, instead of sitting still. An enemy that reaches the bottom of the screen unshot is removed - and does NOT count toward your win condition, since it escaped rather than being defeated.*

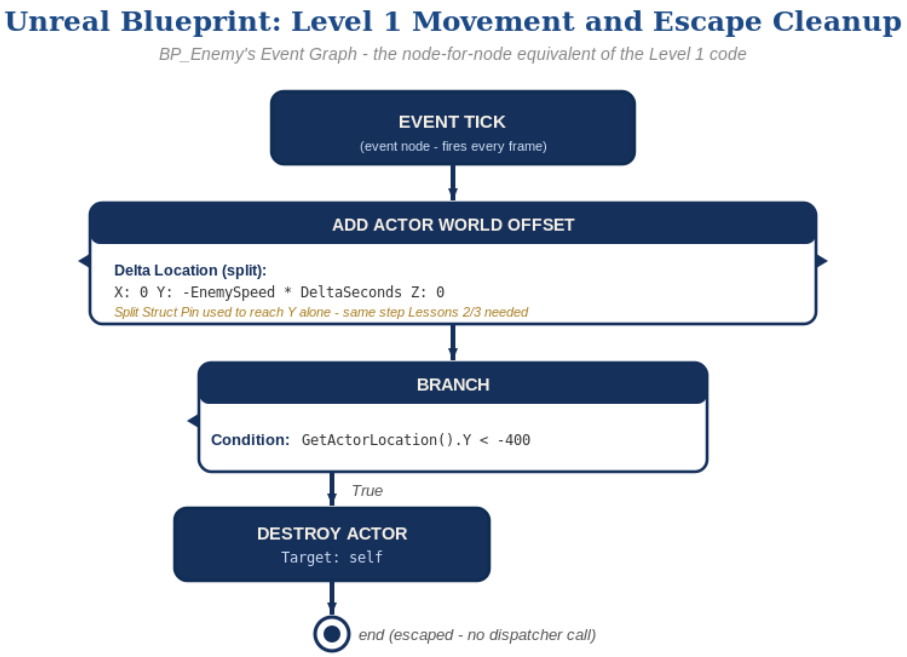



*These two checks look similar - both end with "the enemy is gone" - but only the hit check is allowed to touch enemies_remaining. Attaching the counter to a generic "was destroyed" hook instead of this specific hit check is exactly the bug GameMaker and 001 Game Creator have to fix this lesson.*

*The general escape-vs-hit distinction behind this week's exercise - the same shape in every engine, whatever the exact code or blocks look like.*

2. In BP_Enemy, add a float variable EnemySpeed (default around 50).
3. In BP_Enemy's Event Graph, add an Event Tick node feeding an "Add Actor World Offset" node. Right-click its Delta Location pin and choose "Split Struct Pin" first (it starts collapsed as one Vector pin, and won't accept a value into just Y until it's split), then wire -EnemySpeed * DeltaSeconds into Y (negative, since this course's Y+ convention is "up"). Add a Branch checking its own Y location against an off-screen-at-the-bottom threshold, feeding a plain "Destroy Actor" node on True - not the OnEnemyDestroyed dispatcher.
4. Click Compile, then Play and let the enemies drift downward - shooting one should still work exactly as before, and letting one reach the bottom unshot should just destroy it, with no "You Win!" message.

```
BP_Enemy's Event Graph (new):
Event Tick ->
  Add Actor World Offset(Self, DeltaLocation = (0,
-EnemySpeed*DeltaSeconds, 0))
Event Tick ->
  Branch(GetActorLocation().Y < -400) -> True: Destroy Actor(Self)
  (plain Destroy Actor here - do NOT call OnEnemyDestroyed)
```




*Compare this graph to the overlap event's successful-Cast branch in the Key: only that separate event ever calls OnEnemyDestroyed, which now feeds BP_GameManager's HandleEnemyDestroyed event - Event Tick above never touches it.*

***BP_GameManager was introduced in Lesson 6 - a placed Actor the Level Blueprint holds a reference to, not the Level Blueprint itself.***

*BP_Enemy's Event Tick graph, in Unreal Blueprint - the node-for-node equivalent of the Level 1 code.*

## Alternate Path: C++ (code, optional advanced track)

*The same movement-and-cleanup pattern, and the same escape-vs-hit distinction, work in C++ instead of Blueprint nodes.*

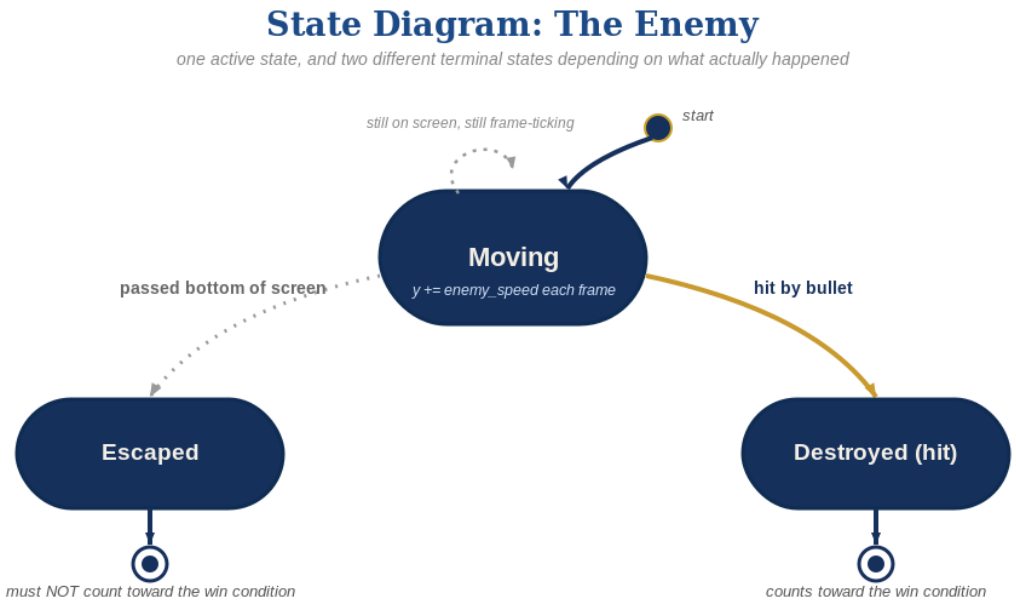
5. In ACppEnemy's Tick(), call AddActorWorldOffset() along -Y, then check GetActorLocation().Y against your off-screen threshold and call a plain Destroy() when past it - not OnEnemyDestroyed.Broadcast().
6. Confirm OnEnemyDestroyed.Broadcast() still only appears inside OnOverlap(), never inside Tick().

```
// ACppEnemy.cpp
void ACppEnemy::Tick(float DeltaSeconds) {
    Super::Tick(DeltaSeconds);
    AddActorWorldOffset(FVector(0.f, -EnemySpeed * DeltaSeconds, 0.f));
    if (GetActorLocation().Y < -400.f) {
        Destroy();
    }
}
```

*This mirrors the Blueprint path's Event Tick and escape Branch - same underlying engine calls, just from code. The escape-vs-hit distinction lives in which function calls Destroy(), the same in both paths.*

## Level 2 - Complete (Alpha/Beta)

*A moving enemy can still be shot and destroyed correctly, and an enemy that escapes off the bottom never triggers "You Win!" by accident.*




*Once an enemy leaves Moving, it never comes back - both destinations are final. The whole trick this lesson teaches is making sure your code can tell these two final states apart, since only one of them is a kill.*

*The enemy's two possible endings this week - only one of them should count as a kill.*

- Shoot a moving enemy to confirm Event ActorBeginOverlap still fires correctly while it's in motion.
- Let at least one enemy escape off the bottom unshot, and confirm "You Win!" does NOT appear - only the successful-Cast branch (a real hit) ever calls OnEnemyDestroyed, and the escape-cleanup Branch never does.

## Level 3 - Refined (Ready to Publish) - pick any that interest you

- Visual: randomize EnemySpeed slightly in BeginPlay (e.g. EnemySpeed = FMath::RandRange(40.f, 70.f) via a Random Float in Range node) so enemies don't all move in perfect lockstep.
- Structure: keep BP_Enemy's escape-cleanup Branch calling only Destroy Actor, with the OnEnemyDestroyed broadcast staying exclusively inside the overlap event's successful-Cast branch.

## If Something's Wrong

- Enemies don't move at all: confirm Event Tick is actually wired to the Add Actor World Offset node, and EnemySpeed is greater than 0.
- Can't wire -EnemySpeed*DeltaSeconds into just the Y of Delta Location - it only shows one combined Vector pin: right-click that pin and choose "Split Struct Pin" to break it into separate X/Y/Z pins first, the same step Lessons 2 and 3 needed.

- "You Win!" appears even when enemies escaped unshot: confirm the escape-cleanup Branch's True case only has a plain Destroy Actor node, with no Call OnEnemyDestroyed nearby.
- Enemies move but never disappear at the bottom: confirm the GetActorLocation().Y < -400 threshold actually matches your level's visible play area.

## Want More?

dev.epicgames.com/community/learning, Student Resource List > Unreal